Ericsson PE Node IND-PE-03 Indiranagar Exchange Hardware/Environmental Fault Resolution SOP

Node: IND-PE-03 | **Vendor:** Ericsson | **Severity:** Major | **Alarm Code:** ALARM_6250

Description

This Standard Operating Procedure (SOP) is designed to address the failure of Ericsson PE Node IND-PE-03 located at the Indiranagar Exchange, which is experiencing a hardware/environmental fault characterized by ASIC memory parity errors, line card crashes, and power supply redundancy loss. This failure domain is specific to the PE role and Ericsson vendor equipment due to the complex interplay of hardware components and environmental factors. The PE node's Line Card (LC) is responsible for providing connectivity to the external network, and its failure can be attributed to issues with the Asynchronous Transfer Mode (ATM) cell bus or the Synchronous Digital Hierarchy (SDH) tributary unit group (TUG). The power supply redundancy loss can be caused by a faulty power supply unit (PSU) or an issue with the power distribution unit (PDU). The ASIC memory parity errors, on the other hand, may be a result of a hardware failure within the ASIC or a software bug. The root cause analysis will involve isolating the specific component or subsystem responsible for the failure to initiate corrective action.

Pre-requisite Checks

- Ensure that the Indiranagar Exchange is accessible and that remote hands are dispatched to the site.
- Verify the status of the redundant link to ensure that it is operational and not contributing to the failure.
- Check the power supply redundancy status to confirm that it is functioning correctly.

Potential Root Causes

Cause	Probability
Line Card (LC) failure due to ATM cell bus or SDH TUG issue	High
Power supply redundancy loss due to faulty PSU or PDU issue	Medium
ASIC memory parity errors due to hardware failure or software bug	Low

Diagnostic Commands & Expected Outputs

```
Command: show lacp neighbor, Expected Output: Neighbor information for lacp, Description: This command will display information about the lacp neighbors, including the port number, neighbor ID, and the lacp state. This will help isolate any issues with the line card's connection to the external network.

Command: show power supply, Expected Output: Power supply information, Description: This command will display information about the power supply units, including their status and any alarms. This will help identify any power supply redundancy issues.
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```
Command: show asic memory errors, Expected Output: ASIC memory error information, Description: This command will display information about any errors detected in the ASIC's memory, including the type of error and the location. This will help identify any hardware failures within the ASIC.
```

Resolution Steps

Step 1: Step: Replace the faulty Line Card (LC), Command: replace lacp lc 1/1/1, Description: Replace the faulty Line Card (LC) with a new one. This will require swapping the LC and ensuring that the new one is properly configured and synchronized with the external network.

Step 2: Step: Replace the faulty power supply unit (PSU), Command: replace power supply 1/1/1, Description: Replace the faulty power supply unit (PSU) with a new one. This will require swapping the PSU and ensuring that the new one is properly configured and synchronized with the power distribution unit (PDU).

Step 3: Step: Reset the ASIC, Command: reset asic 1/1/1, Description: Reset the ASIC to clear any errors or bugs that may be causing the memory parity errors. This will require restarting the node and verifying that the ASIC is functioning correctly.

Post-Check Verification

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Command: show lacp neighbor, Expected Output: Neighbor information for lacp, Description: This command will display information about the lacp neighbors, including the port number, neighbor ID, and the lacp state. This will help verify that the line card is functioning correctly and connected to the external network.

Command: show power supply, Expected Output: Power supply information, Description: This command will display information about the power supply units, including their status and any alarms. This will help verify that the power supply redundancy is functioning correctly.

Command: show asic memory errors, Expected Output: ASIC memory error information, Description: This command will display information about any errors detected in the ASIC's memory, including the type of error and the location. This will help verify that the ASIC is functioning correctly and free of errors.
```

Roll-back Procedures

- Step: Restore the faulty Line Card (LC), Command: restore lacp lc 1/1/1, Description: Restore the faulty Line Card (LC) to its original state. This will require swapping the LC back to its original position and ensuring that it is properly configured and synchronized with the external network.

- Step: Restore the faulty power supply unit (PSU), Command: restore power supply 1/1/1, Description: Restore the faulty power supply unit (PSU) to its original state. This will require swapping the PSU back to its original position and ensuring that it is properly configured and synchronized with the power distribution unit (PDU).

- Step: Reset the ASIC to its original state, Command: reset asic 1/1/1 original, Description: Reset the ASIC to its original state. This will require restarting the node and verifying that the ASIC is functioning correctly and free of errors.